Ikechukwu Iyke-Ogbo

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Education

Covenant University Ogun State, Nigeria
B.Sc. Computer Science Nov 2021 – Sep 2025
Thesis: Development of an Optimal Study Planner Using Hierarchical Task Networks and Constraint Satisfaction
Related Coursework: Software Engineering, Object Oriented Programming, Algorithms and Data Structures, Programming Languages, Graphics and Animation, Human-Computer Interaction, Computer Security, Artificial Intelligence.

Research Interests

Continual Learning, Neural Plasticity, Representation Learning, Learning Dynamics & Neural Network Optimization, Deep Learning Theory, Interpretability of Neural Representations, NLP for Low-Resource African Languages, Applied Machine Learning.

Selected Projects

Temporal Representation Geometry for Neural Plasticity | Independent Research (In Progress) Ongoing

- Developing a research project on whether temporal, layer-wise changes in neural representation geometry (effective rank, CKA, representation drift) predict future trainability in continual learning beyond established dormant-neuron and optimization-based diagnostics.
- Designing a staged experimental plan on MLPs and small CNNs (Permuted MNIST, Split MNIST, Sequential CIFAR-10) to reproduce plasticity loss and test targeted, layer-specific interventions against ReDo and Continual Backpropagation baselines; currently building the theoretical foundation and measurement pipeline.

Speech-to-Speech (S2S) Translation Pipeline | Whisper, Helsinki-NLP, gTTS, Gradio, Google Colab Jul 2026

- Designed and built a full Spanish→English speech-to-speech translation pipeline on Google Colab, chaining Whisper (ASR), a Helsinki-NLP neural machine translation model, and gTTS (speech synthesis) behind a Gradio interface.
- Extended the system with a Yoruba→English research track to evaluate low-resource-language performance, surfacing and documenting failure modes such as ASR hallucination.
- Ran quantitative evaluation of translation and speech-recognition quality (Spanish WER $\approx$ 8%, BLEU $\approx$ 11.5), producing a documented comparison between a high-resource and a low-resource language pair.
- Authored full project documentation covering architecture, evaluation methodology, and results, as a centerpiece portfolio piece for NLP roles.

Financial Insight Application | Independent Project (In Progress) Jul 2026

- Designing an application that analyzes Nigerian bank statements (e.g., GTBank, Access Bank, UBA) to generate spending insights.
- Evaluating architecture trade-offs between fine-tuning and LLM API integration; selected an LLM-API approach using anonymized transaction data to reduce cost and complexity while addressing NDPR data-privacy requirements.

Professional Experience

National Centre for Artificial Intelligence and Robotics (NCAIR) Abuja, Nigeria
Engineering Intern Dec 2025 – Oct 2026

Smart Room Automation System | Infrared Sensors, DHT Sensor, Web Frontend Oct 2026

- Co-built a prototype smart room that automatically cuts power to room appliances when it becomes empty and restores it when someone arrives.
- Used infrared sensors for motion detection and people counting to track room occupancy, and a DHT sensor to monitor room temperature.
- Developed a web frontend that displays live room status, including occupancy and temperature.

Self-Balancing Robot | Arduino Uno, Gyroscope, DC Motors Sep 2026

- Co-built a two-wheeled self-balancing robot on an Arduino Uno that keeps itself upright in real time by reading tilt from a gyroscope and driving the motors in response.
- Integrated the gyroscope and dual DC motors into a closed-loop feedback system, working through sensor readings, motor control, and tuning to achieve stable balance.
- Note: At the end of the day, the robot didn't balance as expected.

FlowPOS — Smart Inventory Surveillance System | ArcFace, YOLOv8, ByteTrack, OpenCV, PyQt5 Jun 2026

- Co-built a cashierless retail system combining ArcFace face recognition, a custom YOLOv8 object detector, and ByteTrack multi-object tracking with automated billing.
- Owned system integration and the PyQt5 desktop UI: built a CameraWorker multithreaded (QThread) architecture connecting the live CV pipeline to the database layer without blocking the interface.
- Trained a custom YOLOv8s snack-detection model (8 classes) on a Roboflow-annotated dataset, reaching ∼88% mAP@50.
- Implemented OpenCV-based tripwire logic to detect item pickup/return events for real-time billing updates.
- Led end-to-end testing and served as presentation lead, defending the system's design (including overfitting mitigations) to a faculty panel.

Image Classification with Transfer Learning | MobileNetV2, Intel Image Classification, Jupyter Jun 2026

- Fine-tuned a MobileNetV2 model on the Intel Image Classification dataset, documenting the full workflow in a Jupyter notebook and accompanying report.

Classic ML & Data Analysis Projects | Python, scikit-learn May 2026

- Built regression and classification pipelines with scikit-learn on the Titanic and house price datasets, covering feature engineering, model selection, and evaluation.
- Contributed to a group analysis of Nigerian hospital performance using NMIS and GRID3 facility data, translating public health data into descriptive insights.

Technical Skills

Languages & Data: Python, NumPy, Pandas, Jupyter, Google Colab, Git/GitHub
ML & Deep Learning: scikit-learn, Transfer Learning, Continual Learning, Model Evaluation (WER, BLEU, mAP), Hugging Face Transformers
Computer Vision: OpenCV, YOLOv8, ArcFace, ByteTrack, Roboflow
NLP & Speech: Whisper (ASR), Neural Machine Translation, gTTS, Gradio
Systems & Embedded: PyQt5, Multithreading (QThread), Arduino, Sensor Integration, Closed-Loop Control
Other: AI Automation, Product Design